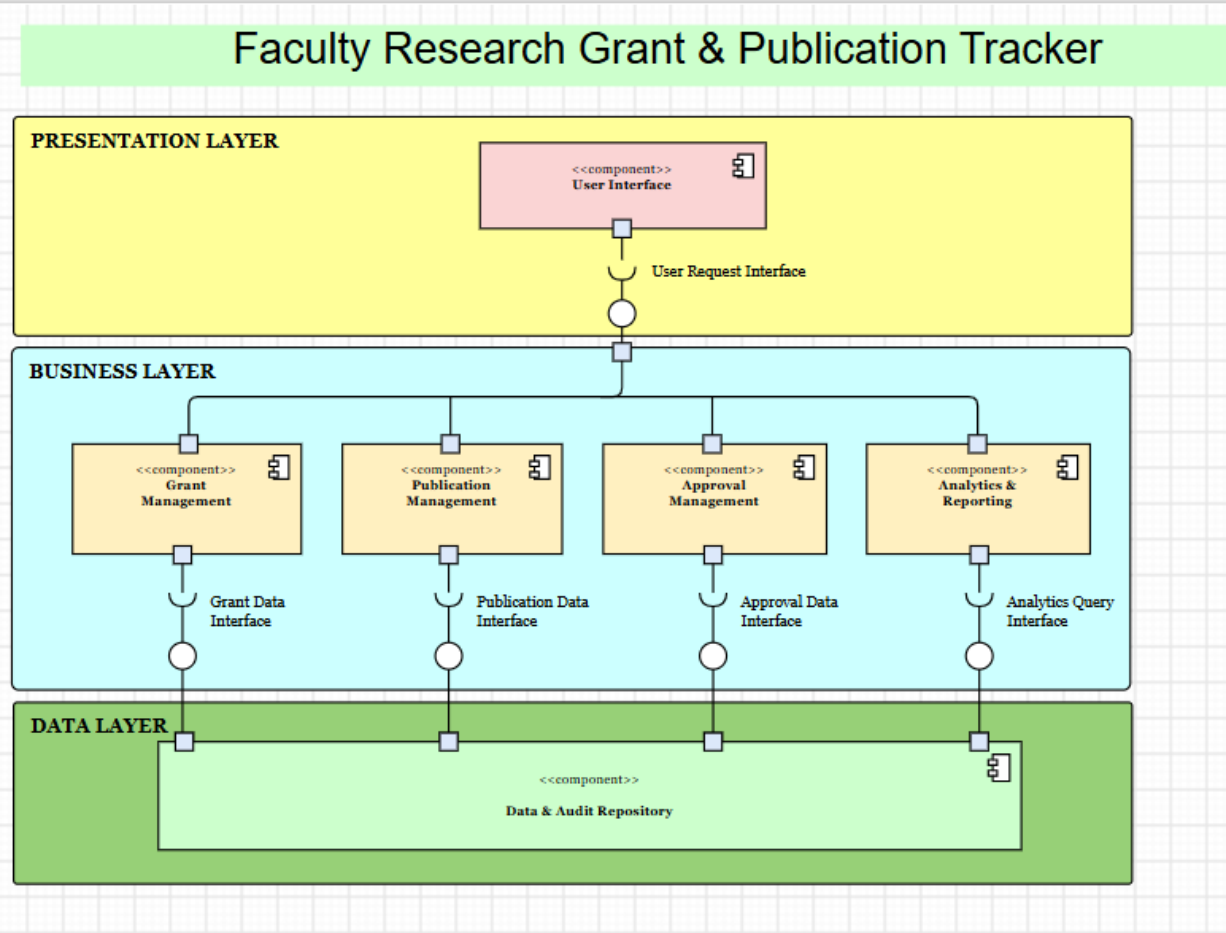
Faculty Research Grant & Publication Tracker

Component Diagram



Written Justification

Architecture Selection

I have opted for Layered Architecture for the Faculty Research Grant & Publication Tracker System. The system

is made up of three main layers: Presentation Layer, Business Layer and Data Layer. The Presentation Layer

would consist of the User Interface; the Business Layer consists of the Grant Management, Publication Management,

Approval Management and Analytics & Reporting, and Data Layer consists of the Data & Audit Repository. This type of

architecture provides a clear separation of concerns and is suitable for conventional applications.

Reason 1 - Separation Of Responsibilities

Layered architecture separates distinct functions of the system into layers. Presentation Layer would be

responsible for communicating with faculty researchers, research managers or any other end-users.

Business Layer would be responsible for grant management, publication management, approving processes,

etc. Data Layer will have complete information on persistent research data.

Reason 2 – Maintainability and Modularity

In its architecture, the system has different independent functional areas, such as grant administration, publication

administration, approval administration, and analytics. By encapsulating them as modular components,

any modifications in one functional area will not affect other components. For example, any changes in the analytics

component can be implemented without any modifications made to the User Interface or any other business components.

The modular architecture also supports easier testing and maintenance of different components.

Security Benefit

Layered architecture allows regulated access to the Data & Audit Repository. The User Interface is not directly connected

to the data; the requests are being processed by several corresponding business components and interfaces. Thus, the

system can protect the sensitive data like grant, publication, approval, and audit related data and establishes an access

control point for the application of authorization and audit restrictions.

Performance Benefit

The separation of analytics and reporting from other business components allows analytics to get grant and expense data

without involving presentation layer in the data processing. Thus, the system is able to minimize extra operations at the use

r interface and allow faster preparation of fund usage and reporting data.